Process & Decision Documentation

Name: Oma Tahir

Role(s): Coder/Implementing idea

Primary responsibility for this work: Used GenAI to implement requirements of this sidequest

Goal of Work Session

I wanted some help from GenAi to achieve the goals of the sidequest. I needed to add a debug screen as well as a toggle option for players.

GenAI Documentation

Date Used: March 20-23 2026

Tool Disclosure: ChatGPT 5.2, Copilot

Purpose of Use: I used GenAI to help me write the prompts to feed into Copilot, and use the built in GenAi Copilot tool to help me implement the necessary changes to get my code to function how it should. I also used ChatGPT 5.2 in the end to help me full in the README file which was missing from the code.

Summary of Interaction: I gave ChatGPT 5.2 the instructions of what I wanted it to do when providing me a input prompt for Copilot. I pasted in the prompts into Copilot that's built into visual studio code, then sent revision prompts where needed. I used ChatGPT in the end by providing it a previous README file and told it to create one for this sidequest.

Human Decision Point(s): The initial changes that were made to the code on visual studios, did not do what I needed them to. I had to scratch those changes and not accept them and re iterate what I was asking for and seeing if those changes did what I needed them to.

Integrity & Verification Note: Making sure I ran the code after sending in my prompts. This allowed me to see that if the changes made where aligned with my task for this side quest.

Scope of GenAI Use: The initial task of providing ChatGPT with what I want it to do was all done by me. Also verifying to see if the code was functioning how I needed it to, and if not then sending in back to revise certain parts of the prompt to make the changes I needed it to.

Limitations or Misfires: It was sometimes misunderstood what I needed it to do, and then would mess up other parts of the code that should have stayed untouched.

Summary of Process (Human + Tool)

Describe what you did, focusing on process rather than outcome. This may include:

- Iteration between drafts and versions
- Testing and revision
- Having to rework what I was asking for to not break the code

Decision Points & Trade-offs

Describe one or two key decisions you made:

- The initial and first prompt did what I needed it to do, but it messed up other aspects of the game like the screen, so I had to scratch that prompt and send in another saying the code needs revisions

Verification & Judgement

Explain how you evaluated whether your decision or change was appropriate:

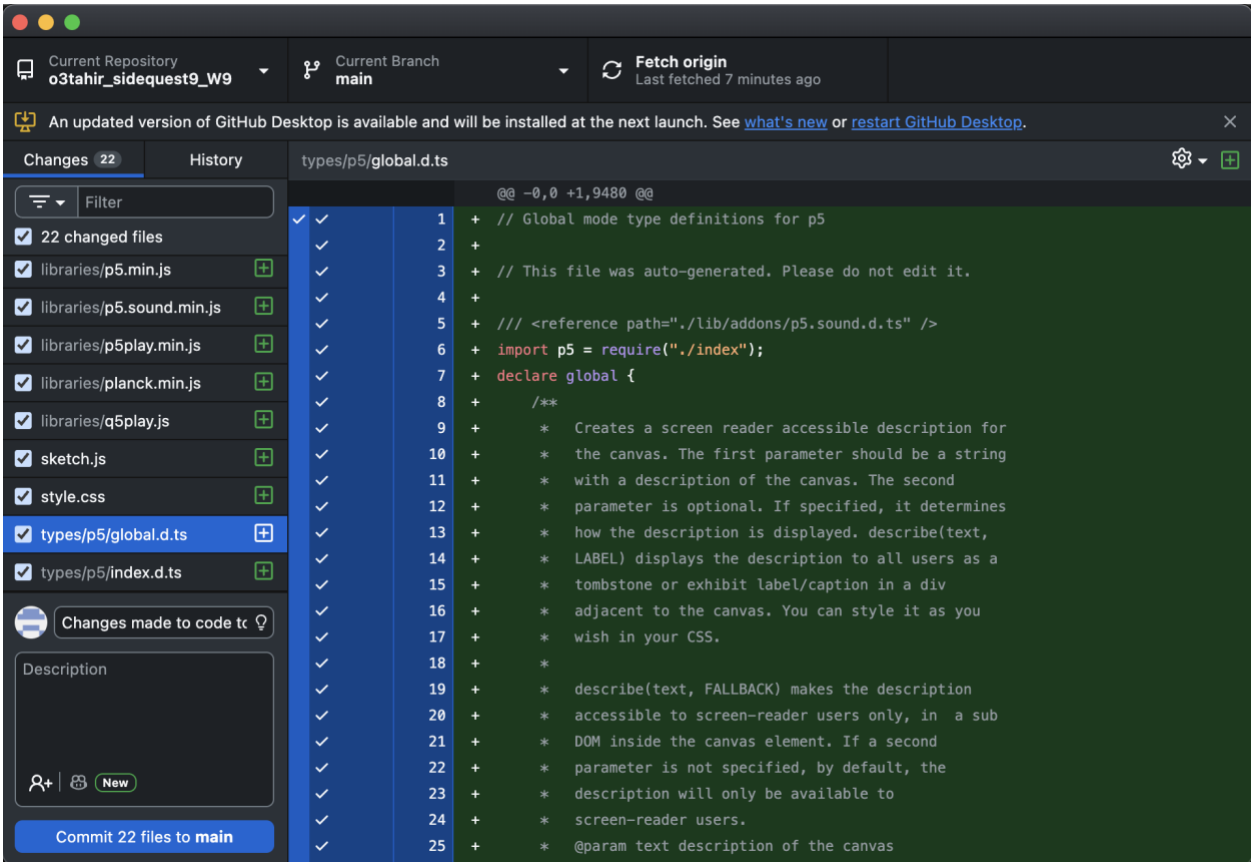
- Re-reading assignment criteria
- Testing the code on live server to see if what I wanted was working properly

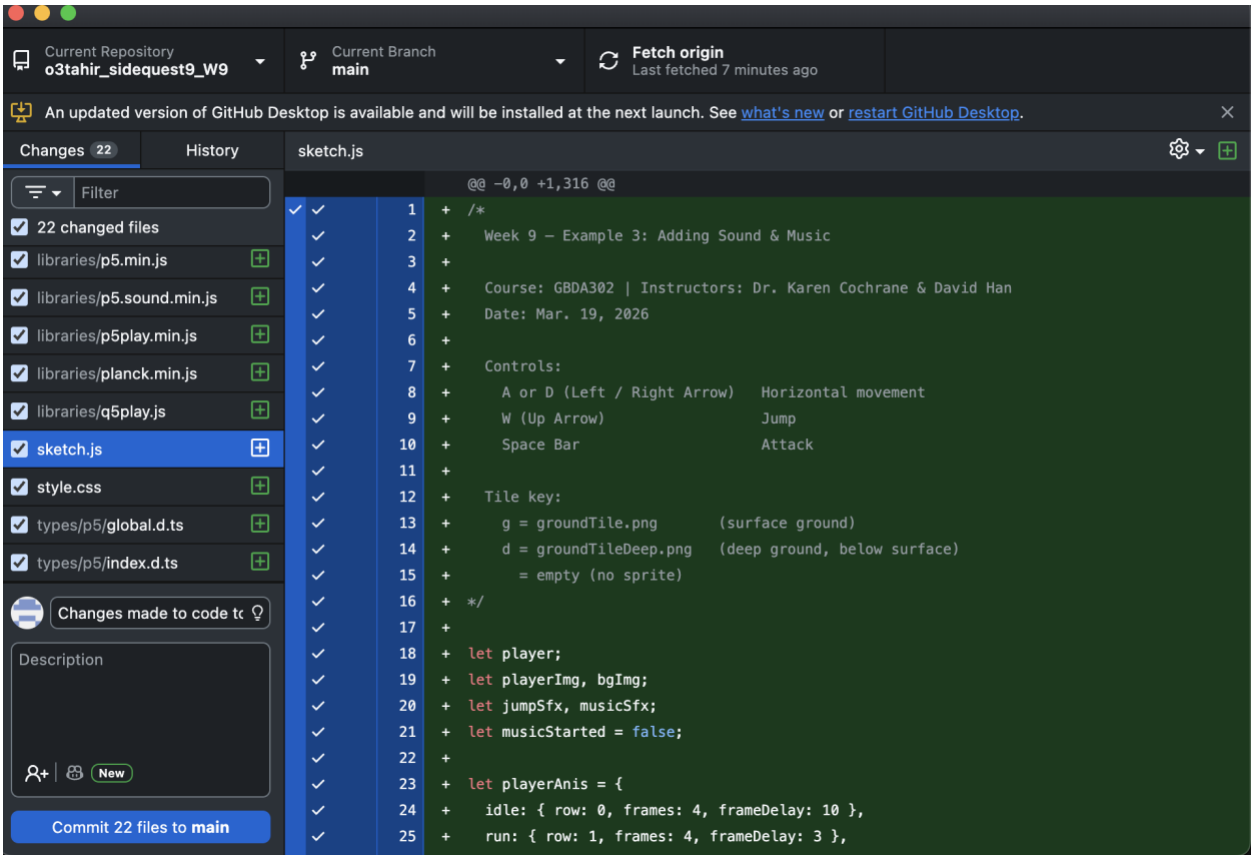
Limitations, Dead Ends, or Open Questions

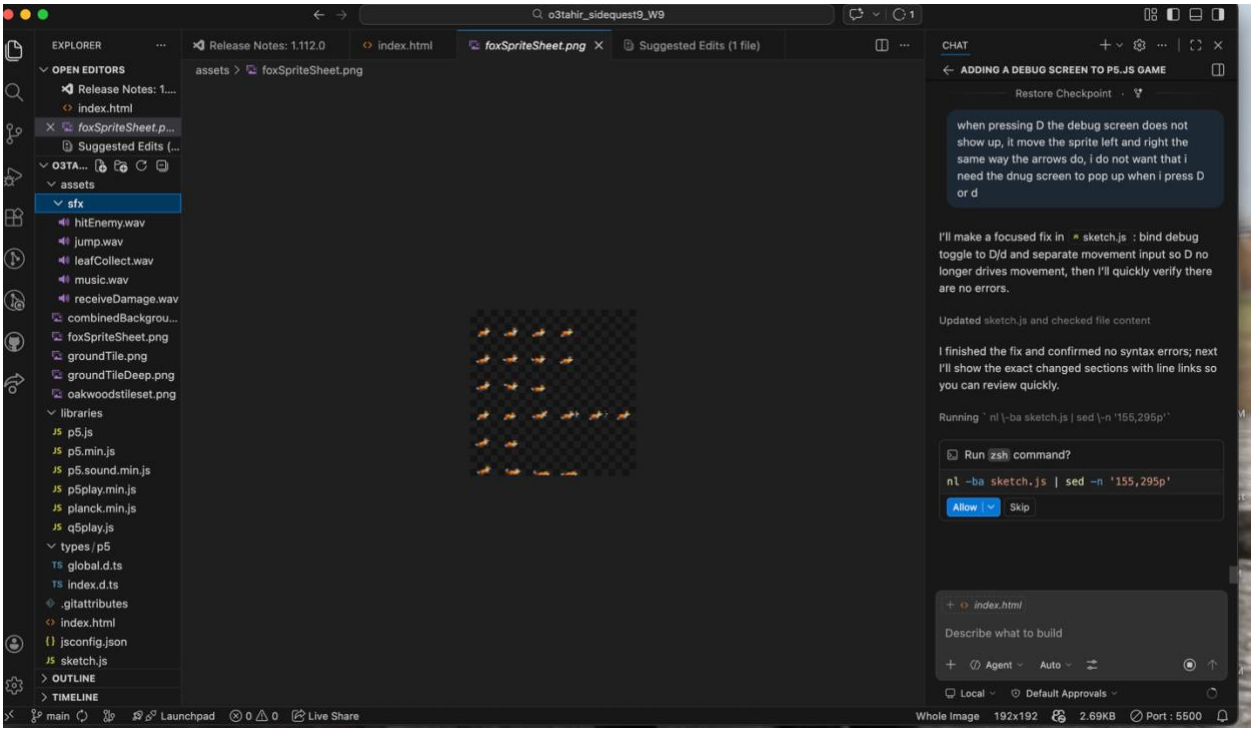
Note anything that:

- N/A

Role-Based Process Evidence




Current Repository
o3tahir_sidequest9_W9
Current Branch
main
Fetch origin
Last fetched 7 minutes ago
An updated version of GitHub Desktop is available and will be installed at the next launch. See what's new or restart GitHub Desktop.
Changes 22
History
sketch.js
Filter
22 changed files
libraries/p5.min.js
libraries/p5.sound.min.js
libraries/p5play.min.js
libraries/planck.min.js
libraries/q5play.js
sketch.js
style.css
types/p5/global.d.ts
types/p5/index.d.ts
Changes made to code to
Description
New
Commit 22 files to main
@@ -0,0 +1,316 @@
+ /*
+ Week 9 — Example 3: Adding Sound & Music
+
+ Course: GBDA302 | Instructors: Dr. Karen Cochrane & David Han
+ Date: Mar. 19, 2026
+
+ Controls:
+ A or D (Left / Right Arrow) Horizontal movement
+ W (Up Arrow) Jump
+ Space Bar Attack
+
+ Tile key:
+ g = groundTile.png (surface ground)
+ d = groundTileDeep.png (deep ground, below surface)
+ = empty (no sprite)
+ */
+
+ let player;
+ let playerImg, bgImg;
+ let jumpSfx, musicSfx;
+ let musicStarted = false;
+
+ let playerAnis = {
+ idle: { row: 0, frames: 4, frameDelay: 10 },
+ run: { row: 1, frames: 4, frameDelay: 3 },



Appendix

Prompt #1

write me a pormot i can put into copiolt to do the following: © Dr. Karen Cochrane and David Han, University of Waterloo, 2026.

Week 9 Side Quest

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• If you completed Side Quest 6, then take your completed Side Quest 6 and

add a debug screen. If you didn't do Side Quest 6, add a debug screen to

Week 9 - Example 3 that allows you to toggle moon gravity on/off, and at

least one other potential feature you might need in developing a game.

Response #1

Help me add a debug screen to my existing p5.js game project.

Requirements:

- Create a debug screen or debug panel that can be opened and closed while the game is running.
- The debug screen must include a toggle for “moon gravity” on/off.
- Also include at least one other useful game development debug feature, such as:
  - showing player coordinates
  - toggling collision boxes / hitboxes
  - showing velocity or speed
  - toggling invincibility / no damage
  - restarting the level
  - showing FPS
- Keep the code simple, clean, and well-commented so I can understand it.
- Use variables and functions that are easy to rename later.
- Make sure the debug features do not break normal gameplay when turned off.

What I want from you:

1. Show me exactly what code to add.
2. Clearly label any NEW code with comments like:

   // NEW: debug mode variables

   // NEW: debug panel drawing

   // NEW: key toggle logic
3. Tell me where each section should go in my file.
4. If needed, create keyboard shortcuts, for example:
   - D = open/close debug screen
   - G = toggle moon gravity
   - H = toggle hitboxes
5. For moon gravity, make gravity weaker than normal so the effect is obvious.
6. Make the debug screen visually simple, like a small overlay box with text showing which debug settings are ON or OFF.

Please write the solution for p5.js and make it fit into an existing game loop with setup(), draw(), and keyPressed() if needed.

Prompt #2

the de bug screen worked but it messed the game itself up

Response #2

The debug screen works, but after adding it, it messed up my game. I need you to FIX the implementation without changing the core gameplay.

Please carefully review the code and do the following:

1. Find what the debug screen changed that is interfering with the game logic, physics, controls, rendering, or collision.

2. Keep all original game behavior the same when debug mode is OFF.

3. Make the debug system isolated, so it only affects the game when a debug toggle is explicitly turned ON.

4. Do NOT rewrite my whole game. Only make the smallest necessary changes.

5. Preserve my existing structure, variables, and gameplay systems as much as possible.

What I need from you:

- Explain what is causing the problem

- Show me exactly what code to change

- Clearly label:

  - code to remove

  - code to replace

  - new code to add

- Add comments beside every fix so I understand why it was changed

Important rules:

- The debug overlay should only draw extra information and should not interrupt normal rendering

- The moon gravity toggle should only change gravity when turned ON, and return to normal when turned OFF

- Any debug keys should not conflict with my gameplay keys

- Debug code should not reset, override, or constantly reassign important gameplay variables every frame unless necessary

- Avoid changing collision, movement, timers, camera, or win/lose logic unless that is the actual bug

- If you create debug variables, keep them separate from gameplay variables where possible

Please check for common problems like:

- gravity being permanently overwritten

- debug keybinds conflicting with movement/action keys

- drawing code being placed in the wrong part of draw()

- state variables being reset every frame

- debug mode accidentally pausing or replacing gameplay logic

- conditionals that stop the main game loop from running

Output format:

1. Brief explanation of the bug

2. Exact fixes

3. Full corrected version of only the affected sections

4. Short summary of why this version is safer

My prompt #3

can you help me come up with stuff i can add to my read mem file which currently has nothing: all i added was This code was takenn from GBDA 302, week 9 example 3. I

Response #3

This project was based on GBDA 302, Week 9 Example 3.

I used the example code as a starting point and then expanded it by adding a debug screen for development and testing purposes. The debug screen allows me to toggle moon gravity on and off, along with other useful debugging features while the game is running.

Features added:

- Debug screen overlay
- Moon gravity toggle
- Additional debug/testing feature(s)
- Keyboard shortcuts for turning debug options on and off

Purpose:

The purpose of this update was to make it easier to test gameplay features and experiment with physics during development without permanently changing the main game experience.

Controls:

- D = open/close debug screen
- G = toggle moon gravity
- [add your other key here] = [say what it does]

Notes:

- The original example code was provided in class as part of GBDA 302 Week 9 Example 3.
- I modified the code to add my own debug functionality.
- The debug features are meant for testing and should not affect normal gameplay when turned off.

Known issues:

- [add this only if needed, for example: “The debug screen still needs some refinement so it does not interfere with gameplay.”]

Author:

Oma Tahir

My prompt #4:

use the old read me as refrence: ## Project Title

GBDA302 Week 4 Example 4: layable Maze (JSON + Level class + Player class)

Authors

Karen Cochrane and David Han. Changes made to code by Oma Tahir using GENAI

Description

Using AI, in the third level the player is now faced with two moving obstecles. They must successfully avoid touching them and getting to the other safe side or else they will have to re-start the level again.

Learning Goals

Learning Goals:

- Loads JSON levels (preload)

- Builds Level objects

- Creates/positions the Player

- Handles input + level switching

It is intentionally light on "details" because those are moved into:

- Level.js (grid + drawing + tile meaning)

- Player.js (position + movement rules)

Assets

N/A

GenAI

The code was written by Dr. Karen Cochrane and David Han but they used GenAI to write the comments.

Oma Tahir used GenAI to make and add changes to the provided code to add obstcles.

Response #4:

Project Title

GBDA 302 Week 9 Side Quest: Debug Screen

Authors

Karen Cochrane and David Han. Changes made to code by Oma Tahir using GenAI.

Description

This project was based on GBDA 302 Week 9 Example 3. Using the provided code as a starting point, a debug screen was added to help with testing and development during gameplay. The debug screen allows the player to toggle moon gravity on and off, along with other helpful debugging features used to check how the game is working.

Learning Goals

Learning Goals:

- Add a debug screen to an existing game
- Create toggles for testing gameplay features
- Use moon gravity as a switchable physics change
- Practice using debug tools during development
- Keep debug features separate from normal gameplay

Assets

N/A

GenAI

The original code was written by Dr. Karen Cochrane and David Han.

Oma Tahir used GenAI to help make and add changes to the provided code in order to create the debug screen, including the moon gravity toggle and other debugging features.

My prompt #5

for asseststheres sounds and sprite sheets which were ncluded in the example code alreayd, i did not add any new pnes, how do i say thta in thew assests part of the sheet

Response #5

Assets

The sound files and sprite sheets used in this project were already included in the original GBDA 302 Week 9 Example 3 code. No additional assets were added.